Which Voter-Behaviour Assumptions Explain Second-Round Results in French Presidential Elections?

Data Acquisition Note • 2 October 2026

1. Group and question

Group: Victor Henz, Edgar Gusching, Louis Bettelli, Lucas Magret Delord, Oscar Moulet.

Main question: Which assumptions about voter behaviour best explain second-round results in French presidential elections (2002–2022)?

Approach: each assumption becomes a simple model that takes a département's first-round results and predicts its second round. We compare predictions with actual results, election by election. The assumptions build on each other, so we can see what each one adds. The best one(s) are then used for a **simulation** (not a prediction) of hypothetical 2027 finalist pairs.

2. What exactly we estimate

One observation = one département in one election: first-round shares (each bloc, blank/null, abstention) and second-round shares (finalist A, finalist B, blank/null, abstention), all over **registered voters** so abstention is part of the model. **A transfer** = the share of a first-round group (including abstainers) that goes to each second-round option. Each assumption gives a different transfer matrix β:

$$\text{Round2}_k(d) = \sum_j \beta_{jk}\,\text{Round1}_j(d), \qquad \sum_k \beta_{jk} = 1, \quad \beta_{jk} \geq 0$$

We only observe aggregates, so results are *estimated aggregate transfer patterns*, never statements about individual voters.

3. The assumptions we test

Based on the advice of Prof. Magdaléna (voting theory), plus two additions of ours (marked *):

- **H0 – Naive baseline*:** voters of eliminated candidates split like the rest of the electorate. Without it, we cannot tell whether the other assumptions add anything.
- **H1 – Ordinal axis:** candidates are ranked left to right (order only); each voter goes to the closest finalist in that order.
- **H2 – Ideological distances:** same idea with real distances, in 1D (left–right) and 2D (economic + social). *If both finalists are too far away, the voter abstains or votes blank.
- **H3 – Noise:** a share p of voters does not follow the closest-candidate rule; p is calibrated on past elections.
- **H4 – Strategic voting (vote utile):** a first-round vote is not always the true first choice, so real preferences are more spread out than first-round scores.
- **H5 – Transfers learned by political family:** no fixed rule; transfers are learned from preference data, with candidates replaced by labels (far left, ecologist, centre…) so they carry over between elections.

Evaluation (same for all): error between predicted and actual second round, by département and nationally. We fit on some elections and test on others, so an assumption does not win just by having more parameters. H3 and H5 involve fitting parameters; we will confirm the method with you first.

4. Data, how we acquire it, and which assumption uses it

Source	What we take	How we acquire it	Used for
Ministry of the Interior raw results	Presidential results 2002–2022, both rounds, by polling station	Script downloads original files per election; we harmonise formats ourselves	**All:** first round = input, second round = ground truth; H3 calibration
Consolidated election dataset	Same results, already harmonised	Downloaded once	Checking our own cleaning only
Chapel Hill Expert Survey	Party scores: general, economic and social left–right; party families	Download, filter France, map each candidate to a party, documented	H1 (order), H2 (distances), H5 (labels)
Polling companies' axes, via Wikipedia	Candidate order on the left–right axis; hypothetical 2027 matchups	Scraping script, saved with collection date	H1 (alternative order); choosing and comparing 2027 simulations
2027 polls			
Political Compass 2022, CEVIPOF surveys	2D candidate positions; candidates' stances on issues	Manual extraction, documented	H2 (alternative distances, robustness check)
Voter Autrement 2017	Approval votes for all candidates, with each participant's official vote	Download from Zenodo	H4 (official vote vs. true preferences), H5 (learning transfers)

All scripts, download dates and file checksums are kept in the repository README so anyone can rebuild the raw data.

5. Where we are now

☒ Question and assumptions defined after your feedback and Prof. Magdaléna's advice ☒ Data sources identified and checked as open and accessible
☐ Acquisition script for 2017 and 2022 raw results (first priority) ☐ Extend to 2002, 2007 and 2012
☐ Candidate-to-party table built from CHES (H1, H2, H5) ☐ Scraper for 2027 polling tables

6. Limits and changes since the proposal

Known limits: aggregate data only (no individual behaviour); expert left–right scores are a modelling choice; small candidates may be missing from CHES and will be placed by bloc; H4 and H5 rely on 2017 preference data only and are treated as extensions; polling-station codes and commune boundaries change between elections; 2027 results are a simulation, not a forecast.

Changes after feedback: dropped the newspaper-mentions data (it did not explain any variable in our question); removed the planned shrinkage for small départements (we will decide after looking at the data); dropped socioeconomic indicators to keep the scope focused; reframed the project as a comparison of several behaviour assumptions, with 2027 as a simulation, and defined precisely what one observation and one transfer are.